

EMPLOYEE DATASET VISUALIZATION

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Course Code: DSA0602

Course Name: Data Handling And Visualization

Dataset Overview

head(employee)

str(employee)

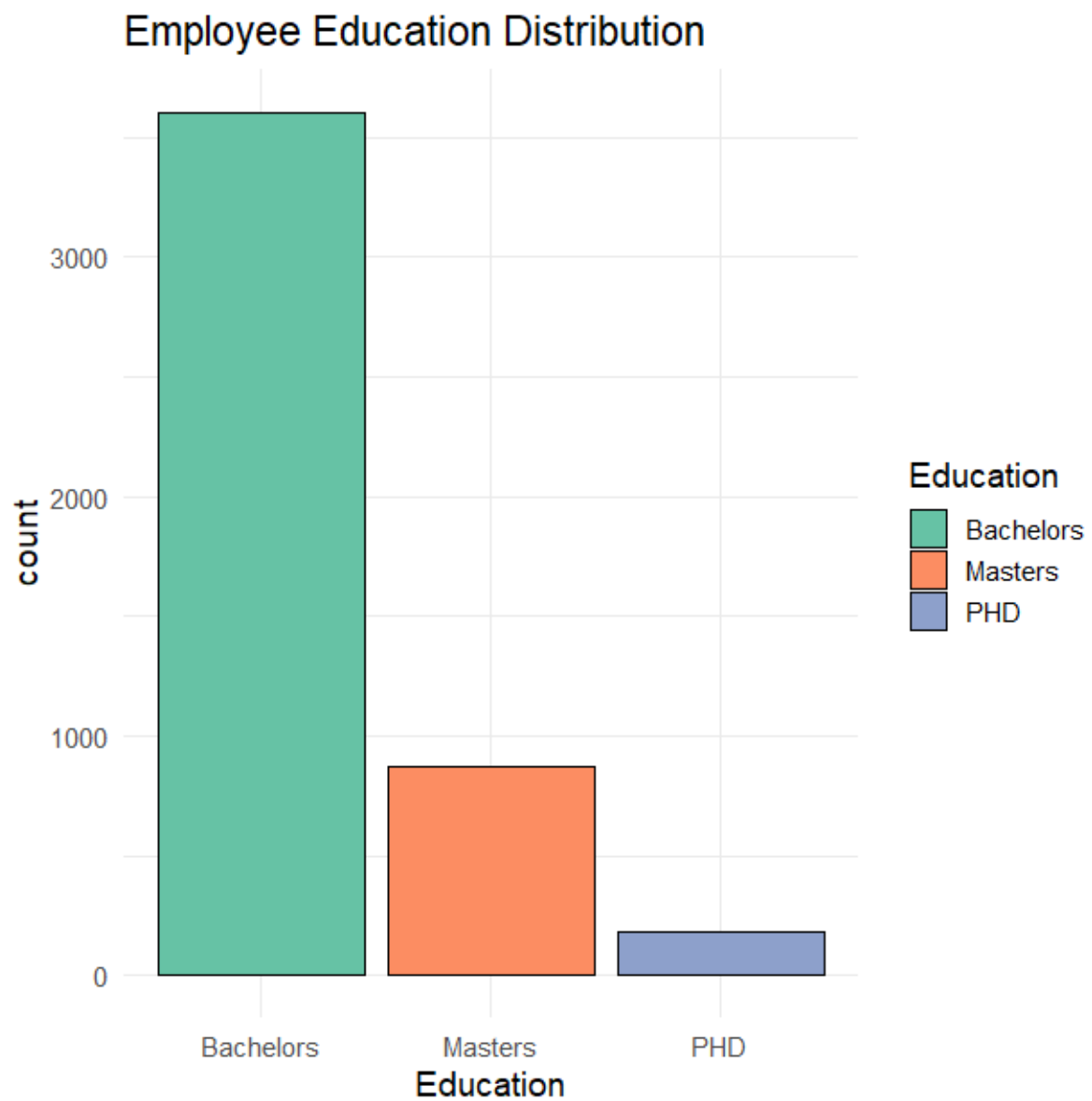
summary(employee)

View(employee)

	Education	JoiningYear	City	PaymentTier	Age	Gender	EverBenched
1	Bachelors	2017	Bangalore	3	34	Male	No
2	Bachelors	2013	Pune	1	28	Female	No
3	Bachelors	2014	New Delhi	3	38	Female	No
4	Masters	2016	Bangalore	3	27	Male	No
5	Masters	2017	Pune	3	24	Male	Yes
6	Bachelors	2016	Bangalore	3	22	Male	No
7	Bachelors	2015	New Delhi	3	38	Male	No
8	Bachelors	2016	Bangalore	3	34	Female	No
9	Bachelors	2016	Pune	3	23	Male	No
10	Masters	2017	New Delhi	2	37	Male	No
11	Masters	2012	Bangalore	3	27	Male	No
12	Bachelors	2016	Pune	3	34	Male	No
13	Bachelors	2018	Pune	3	32	Male	Yes
14	Bachelors	2016	Bangalore	3	39	Male	No
15	Bachelors	2012	Bangalore	3	37	Male	No
16	Bachelors	2017	Bangalore	1	29	Male	No
17	Bachelors	2014	Bangalore	3	34	Female	No
18	Bachelors	2014	Pune	3	34	Male	No
19	Bachelors	2015	Pune	2	30	Female	No

1. Bar Chart

```
ggplot(employee, aes(x = Education, fill = Education)) +  
  geom_bar(color = "black") +  
  scale_fill_brewer(palette = "Set2") +  
  labs(title = "Employee Education Distribution") +  
  theme_minimal(base_size = 14)
```

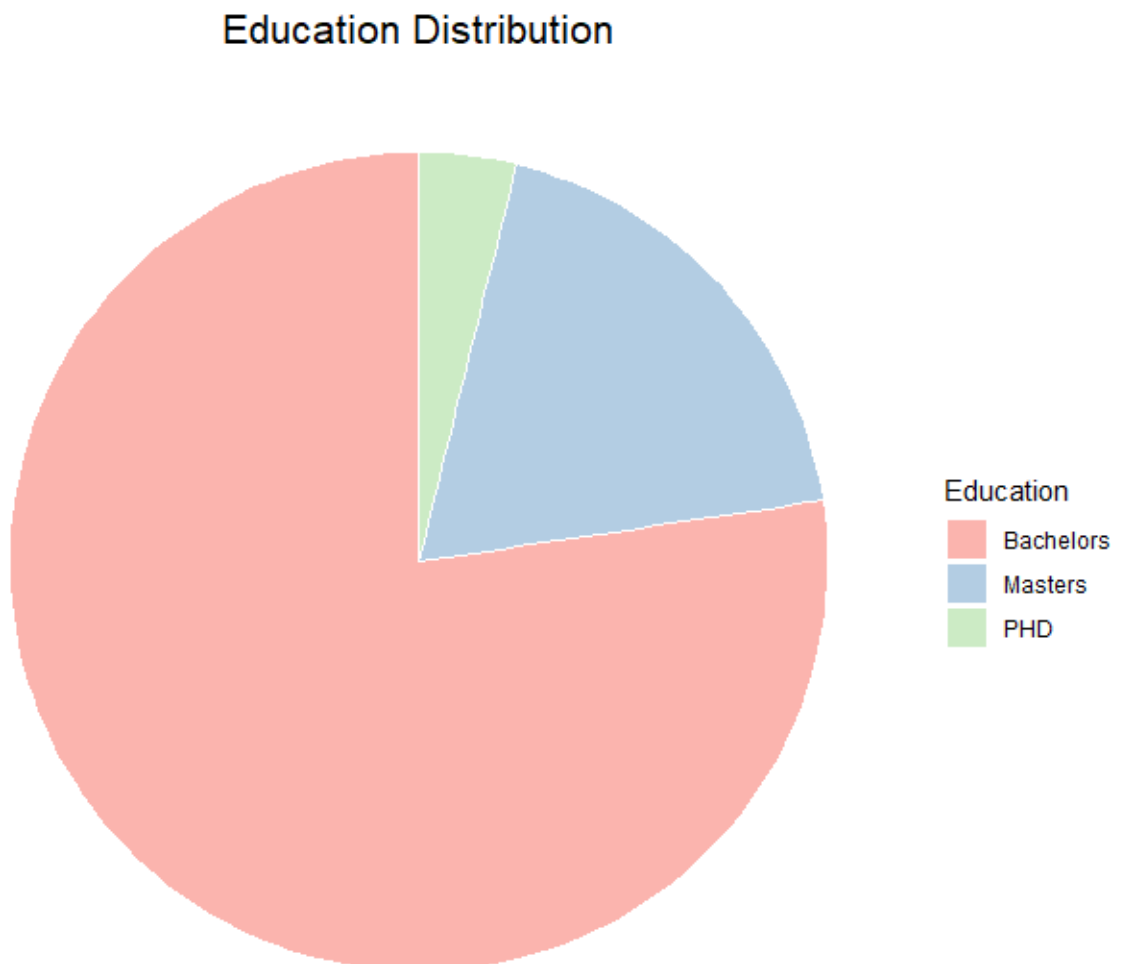


2. Pie Chart

```
library(dplyr)
```

```
edu <- employee %>%  
  count(Education)
```

```
ggplot(edu, aes(x = "", y = n, fill = Education)) +  
  geom_bar(stat = "identity", width = 1, color = "white") +  
  coord_polar("y") +  
  scale_fill_brewer(palette = "Pastel1") +  
  labs(title = "Education Distribution") +  
  theme_void() +  
  theme(plot.title = element_text(hjust = 0.5, size = 16))
```



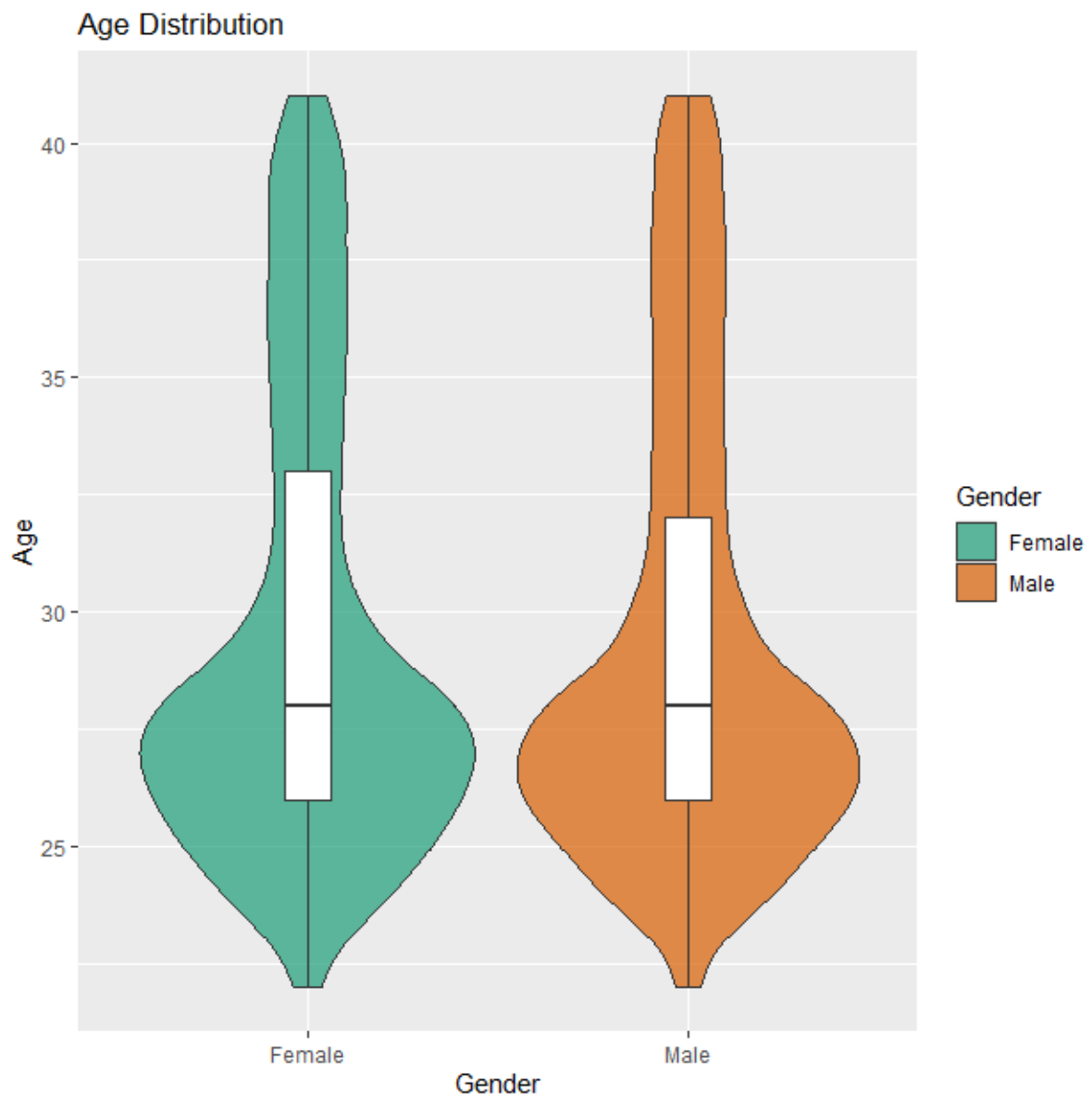
3. Box Plot

```
ggplot(employee, aes(x = Gender, y = Age, fill = Gender)) +  
  geom_boxplot() +  
  scale_fill_manual(values = c("#FF69B4", "#00BFFF")) +  
  labs(title = "Age Distribution by Gender") +  
  theme_classic(base_size = 14)
```



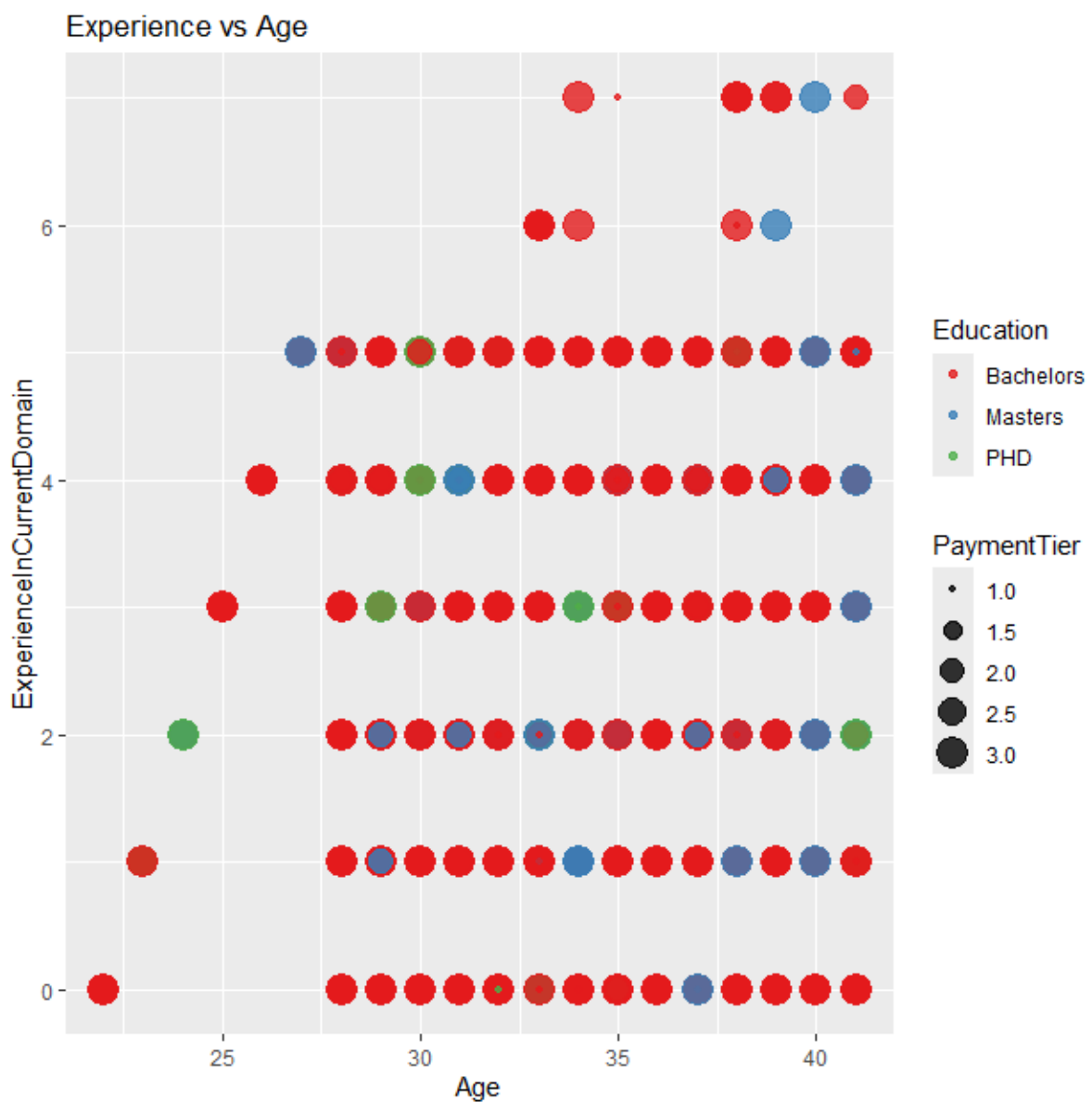
4. Violin Plot with Boxplot

```
ggplot(employee, aes(Gender, Age, fill = Gender)) +  
  geom_violin(alpha = 0.7) +  
  geom_boxplot(width = 0.12, fill = "white") +  
  scale_fill_brewer(palette = "Dark2") +  
  labs(title = "Age Distribution")
```



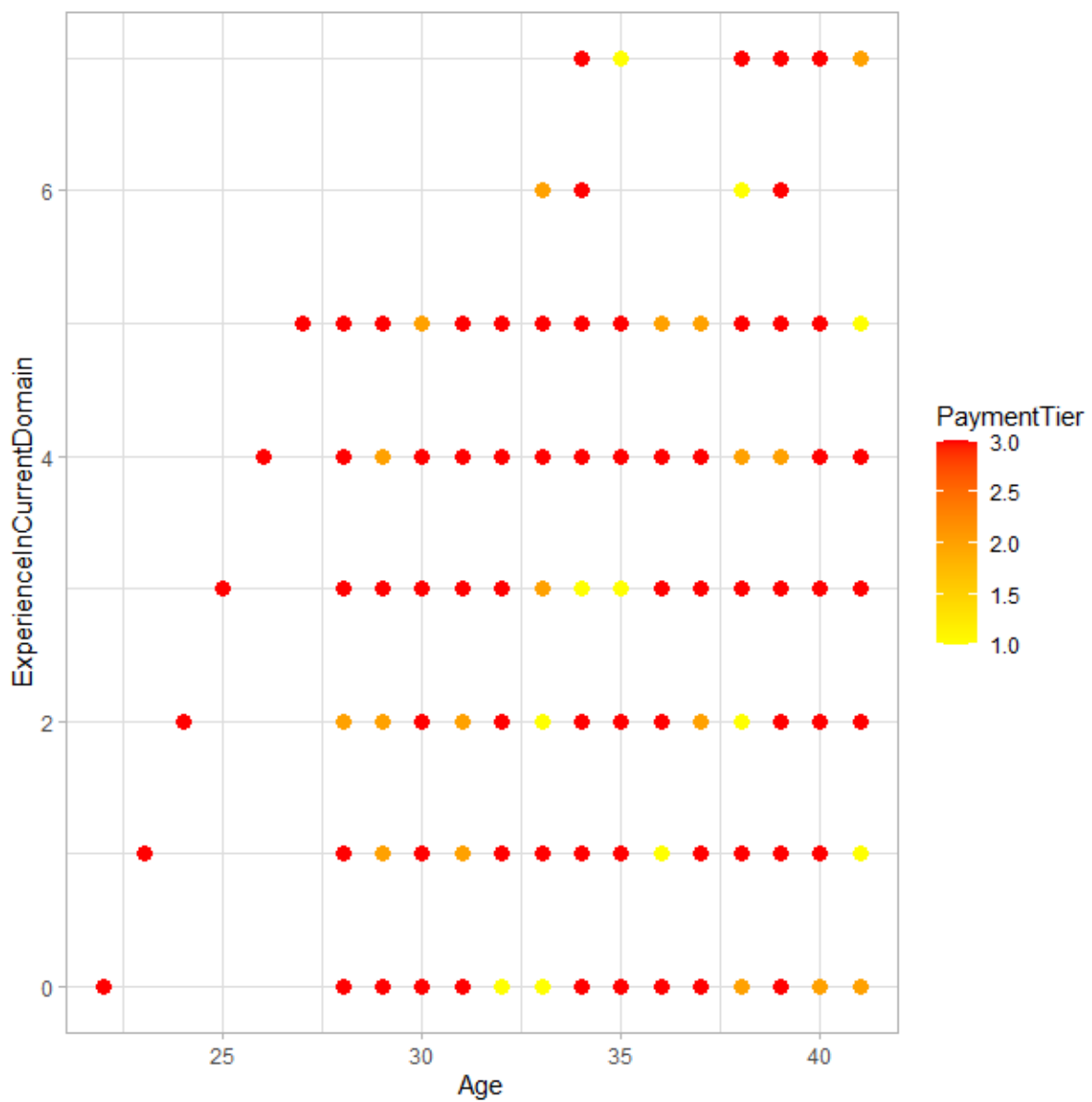
5. Bubble Chart

```
ggplot(employee,  
  aes(Age,  
    ExperienceInCurrentDomain,  
    size = PaymentTier,  
    color = Education)) +  
geom_point(alpha = 0.8) +  
scale_color_brewer(palette = "Set1") +  
labs(title = "Experience vs Age")
```



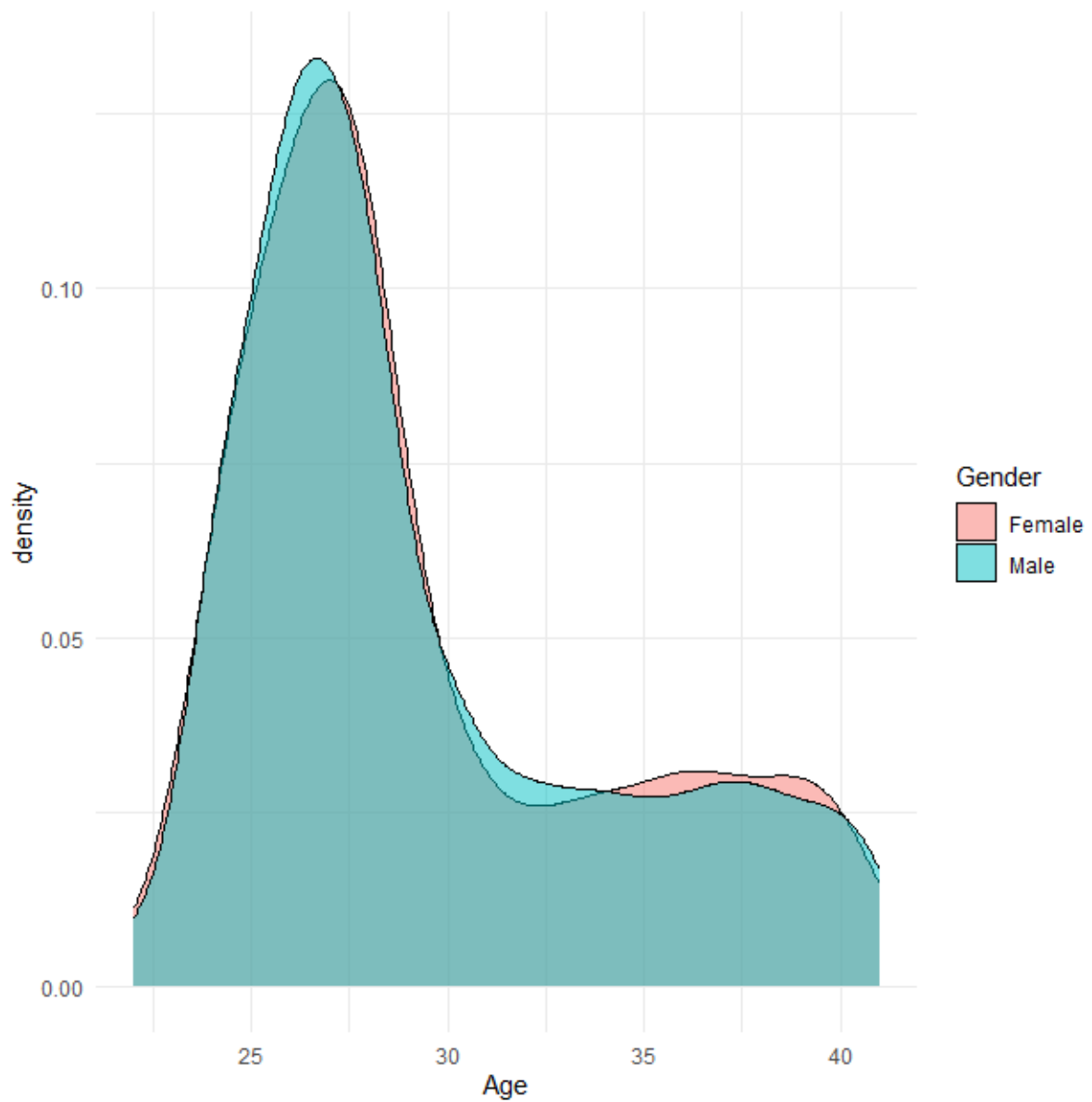
6. Gradient Scatter Plot

```
ggplot(employee,  
  aes(Age,  
    ExperienceInCurrentDomain,  
    color = PaymentTier)) +  
geom_point(size = 3) +  
scale_color_gradient(low = "yellow",  
  high = "red") +  
theme_light()
```



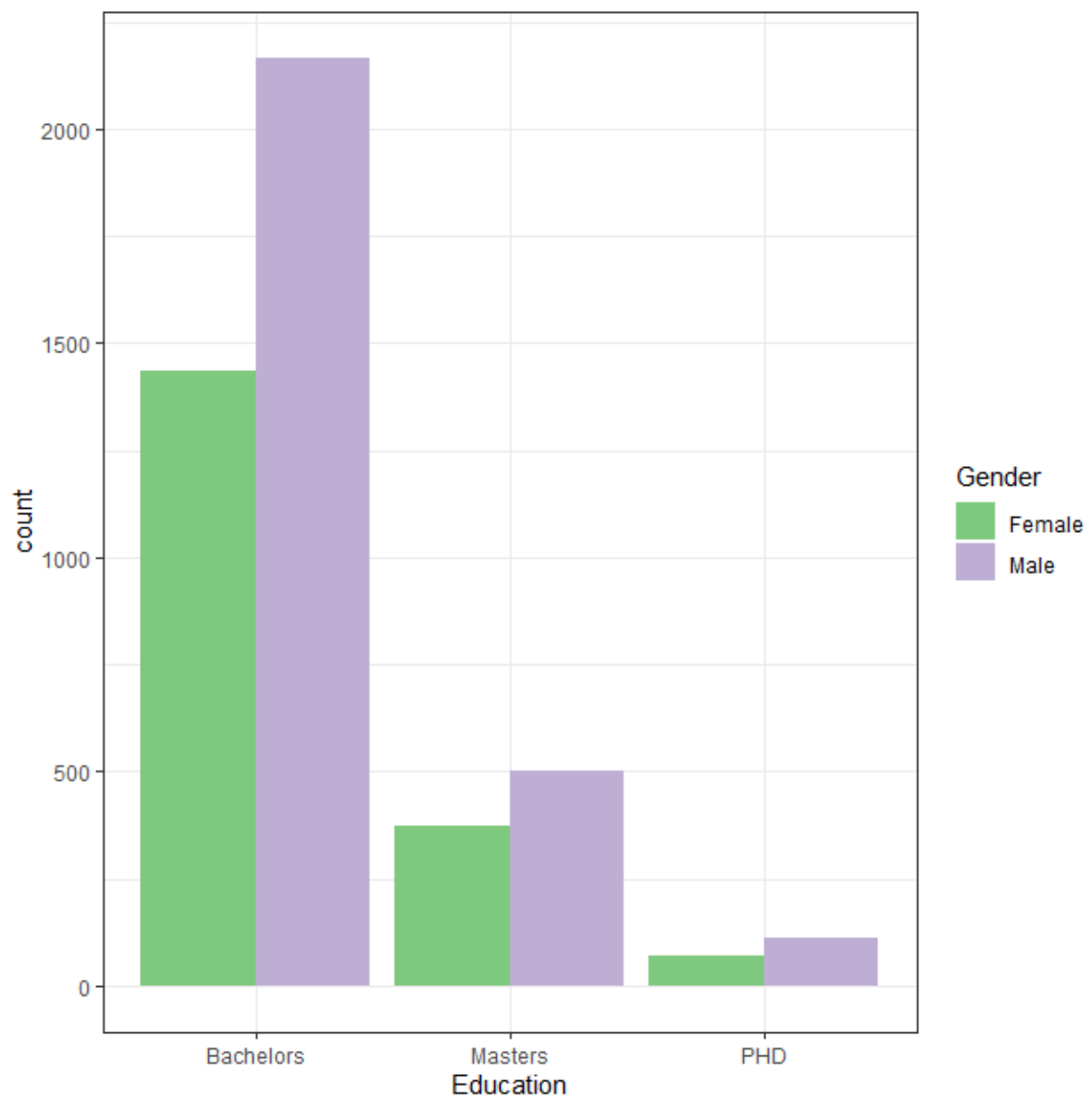
7. Density Plot

```
ggplot(employee,  
  aes(Age,  
    fill = Gender)) +  
  geom_density(alpha = 0.5) +  
  scale_fill_manual(values = c("#F8766D","#00BFC4")) +  
  theme_minimal()
```



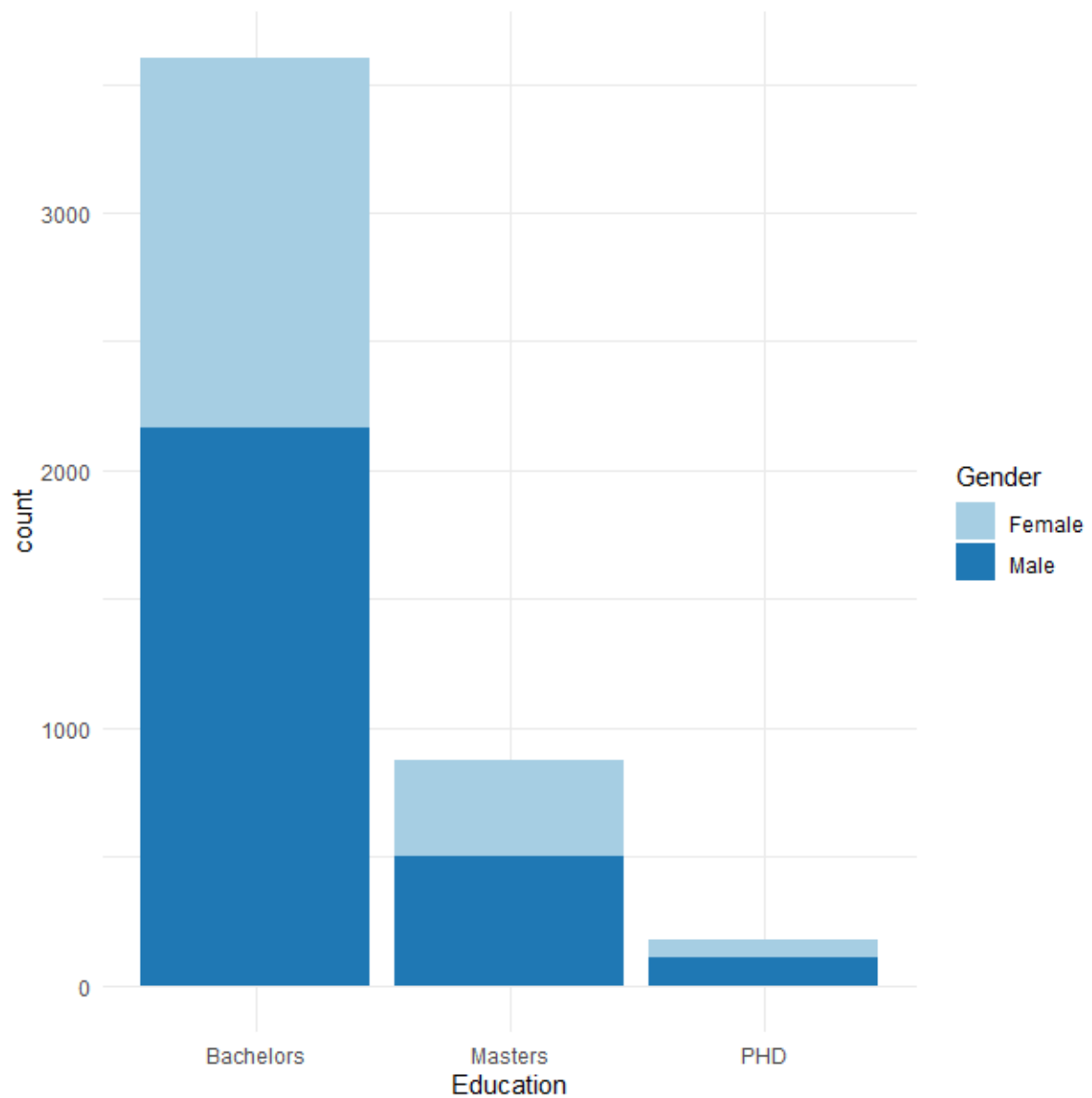
8. Grouped Bar Chart

```
ggplot(employee,  
  aes(Education,  
    fill = Gender)) +  
  geom_bar(position = "dodge") +  
  scale_fill_brewer(palette = "Accent") +  
  theme_bw()
```



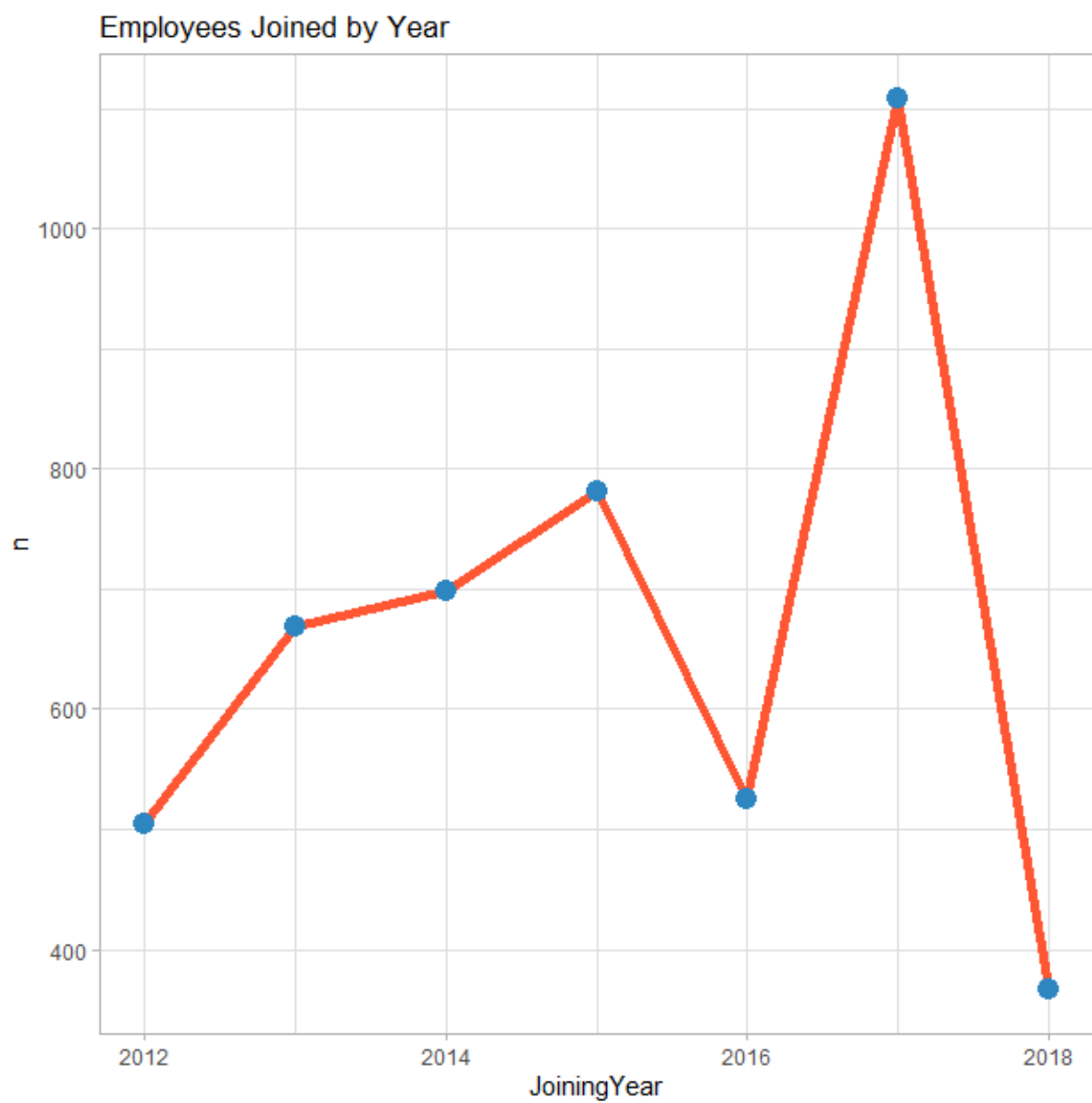
9. Stacked Bar Chart

```
ggplot(employee,  
  aes(Education,  
    fill = Gender)) +  
geom_bar() +  
scale_fill_brewer(palette = "Paired") +  
theme_minimal()
```



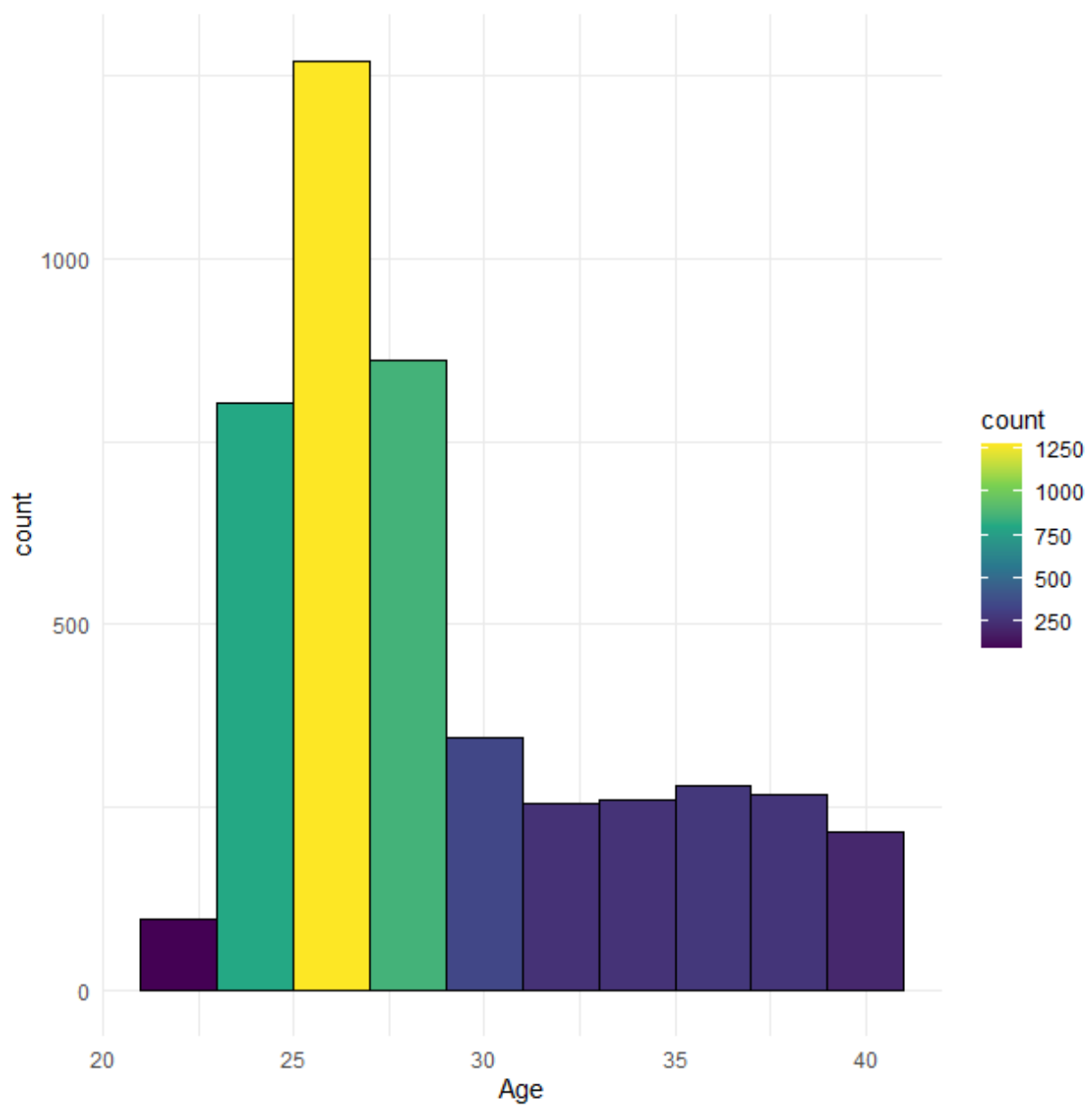
10. Smooth Line Chart

```
employee %>%  
  count(JoiningYear) %>%  
  ggplot(aes(JoiningYear, n)) +  
  geom_line(color = "#FF5733", linewidth = 2) +  
  geom_point(size = 4, color = "#2E86C1") +  
  theme_light() +  
  labs(title = "Employees Joined by Year")
```



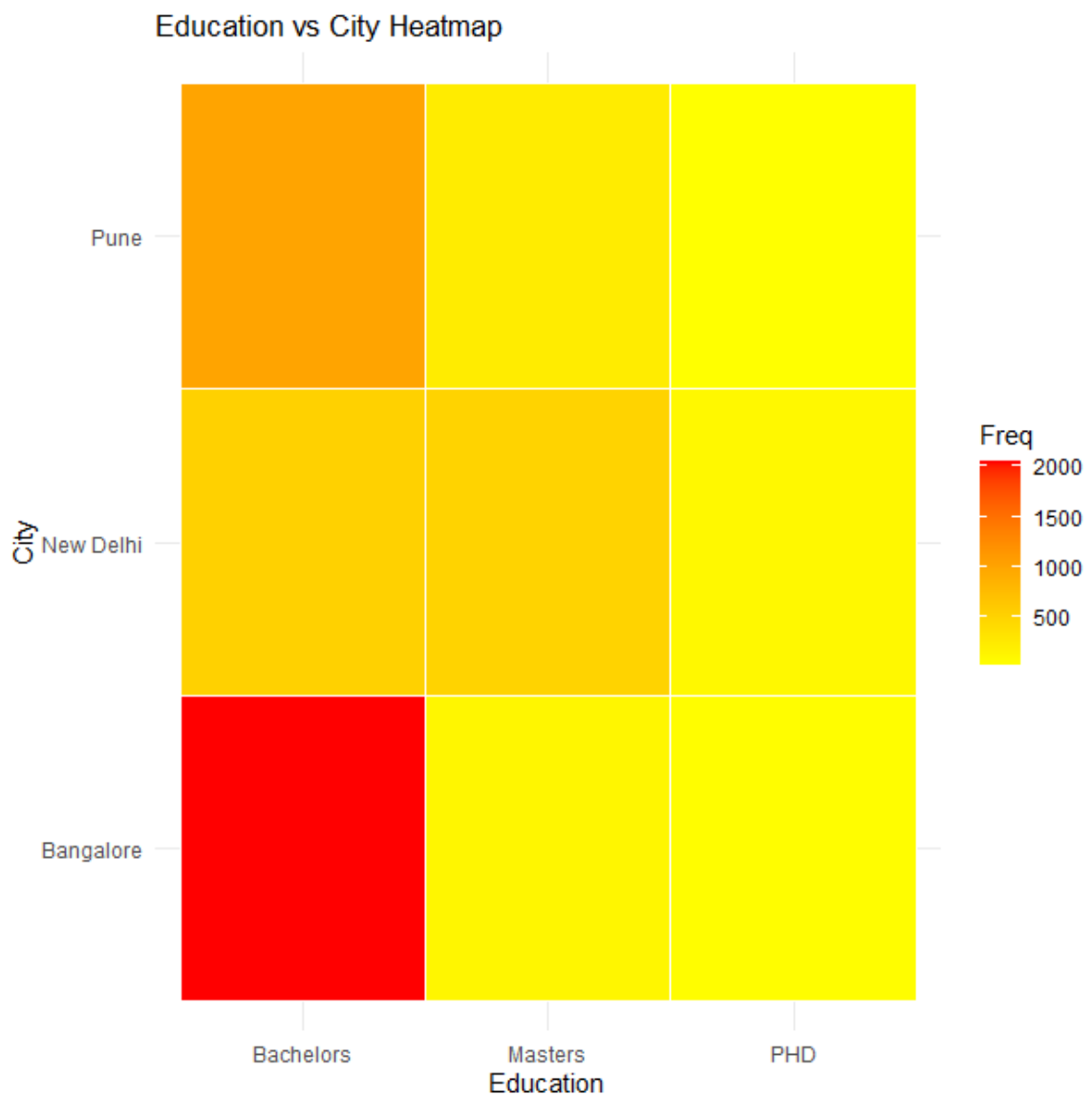
11. Histogram

```
ggplot(employee,  
  aes(Age,  
    fill = after_stat(count))) +  
geom_histogram(binwidth = 2,  
  color = "black") +  
scale_fill_viridis_c() +  
theme_minimal()
```



12. Heatmap

```
library(ggplot2)
tbl <- as.data.frame(table(employee$Education,
                           employee$City))
ggplot(tbl,
       aes(Var1,
           Var2,
           fill = Freq)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "yellow",
                     high = "red") +
  labs(title = "Education vs City Heatmap",
       x = "Education",
       y = "City") +
  theme_minimal()
```

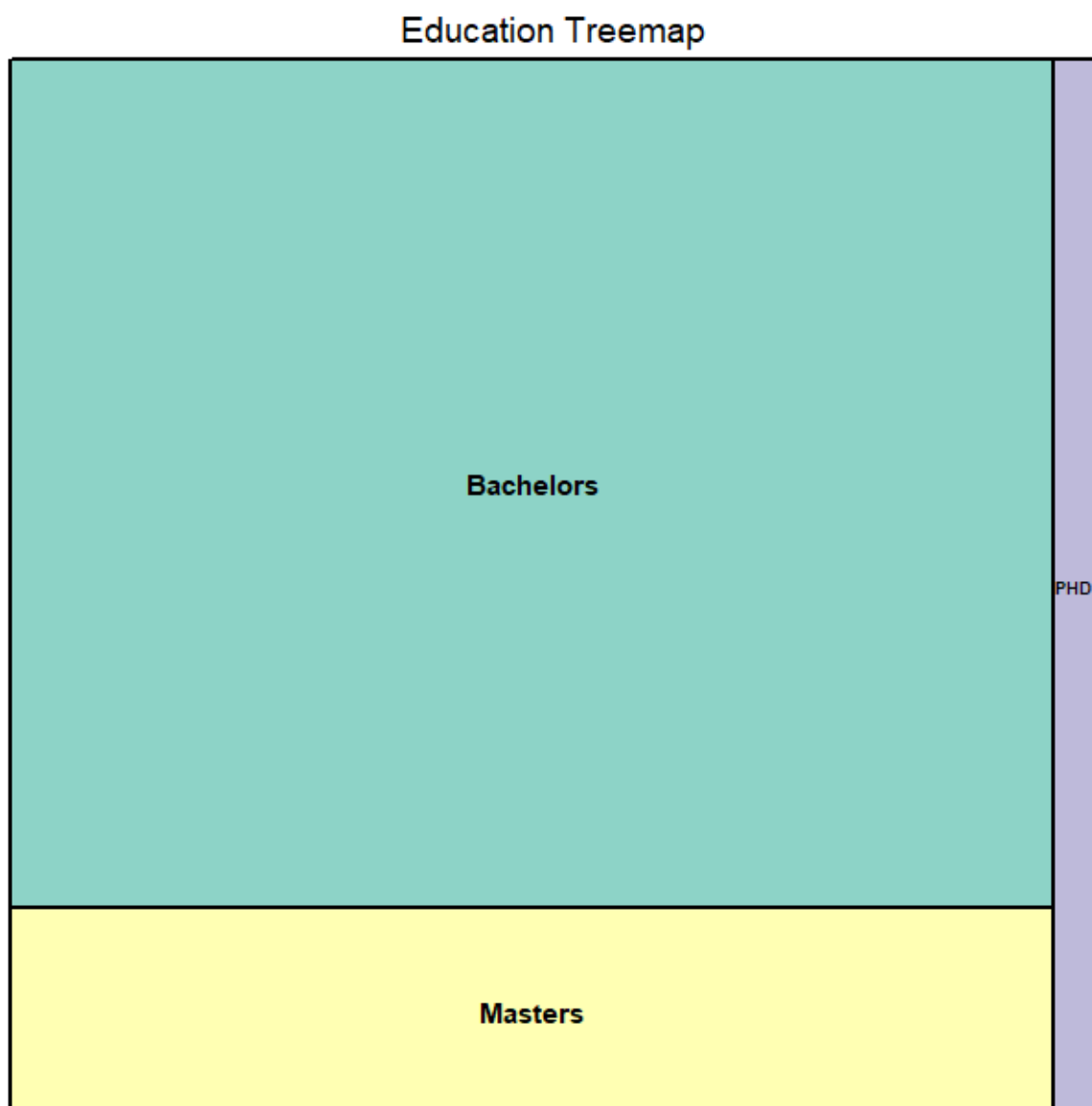


13. Treemap

```
library(treemap)
```

```
tree <- employee %>%  
  count(Education)
```

```
treemap(tree,  
  index = "Education",  
  vSize = "n",  
  palette = "Set3",  
  title = "Education Treemap")
```

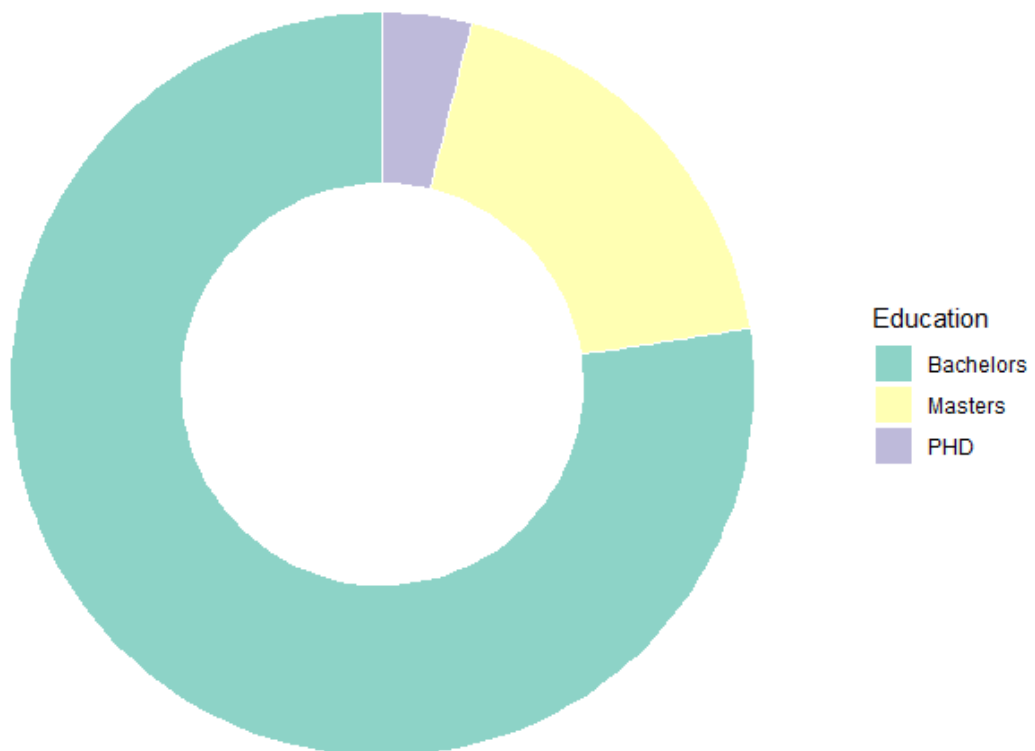


14. Donut Chart

```
edu <- employee %>%  
  count(Education)
```

```
ggplot(edu, aes(x = 2, y = n, fill = Education)) +  
  geom_col(color = "white") +  
  coord_polar(theta = "y") +  
  xlim(0.5, 2.5) +  
  scale_fill_brewer(palette = "Set3") +  
  theme_void() +  
  labs(title = "Education Donut Chart")
```

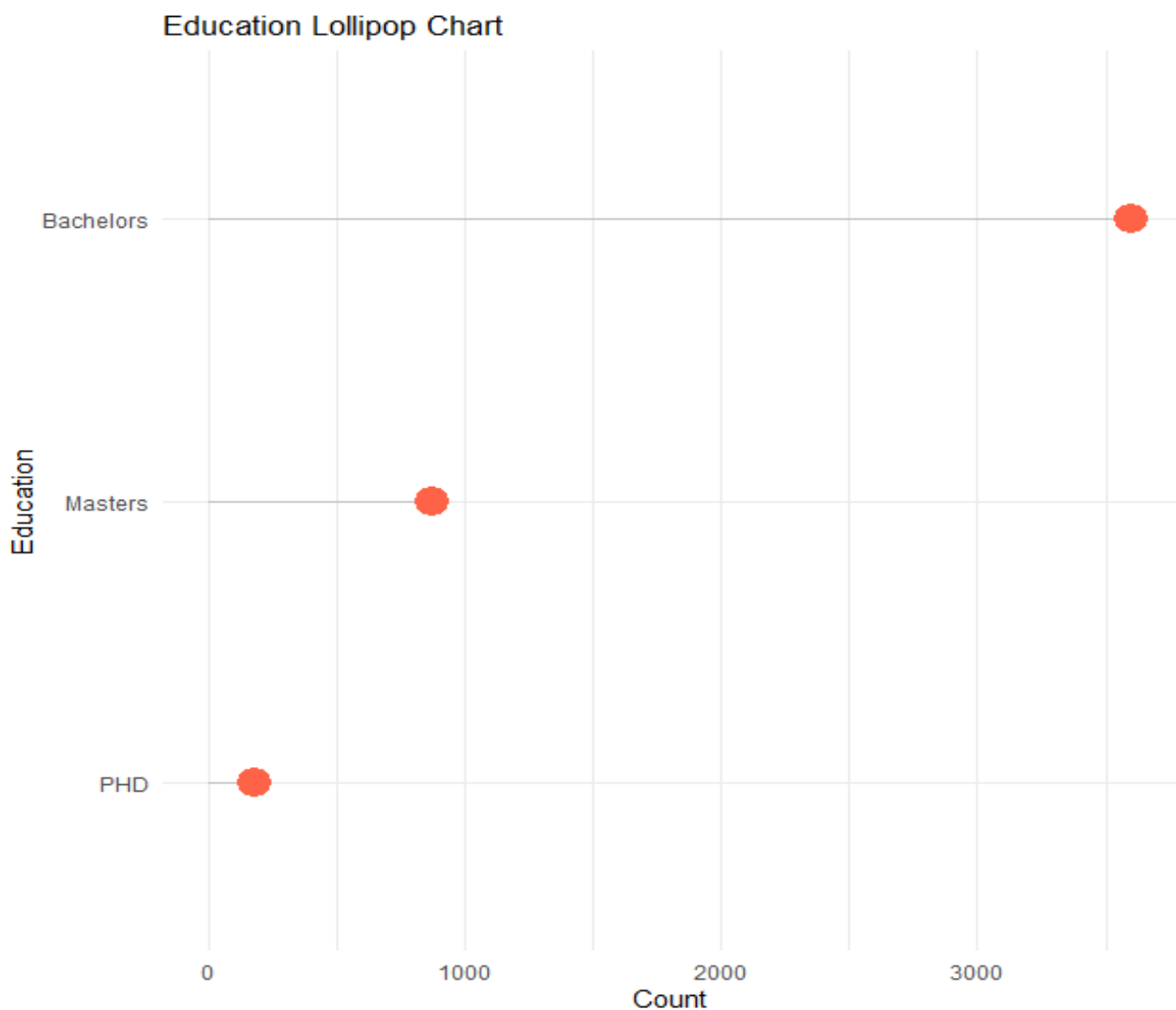
Education Donut Chart



15. Lollipop Chart

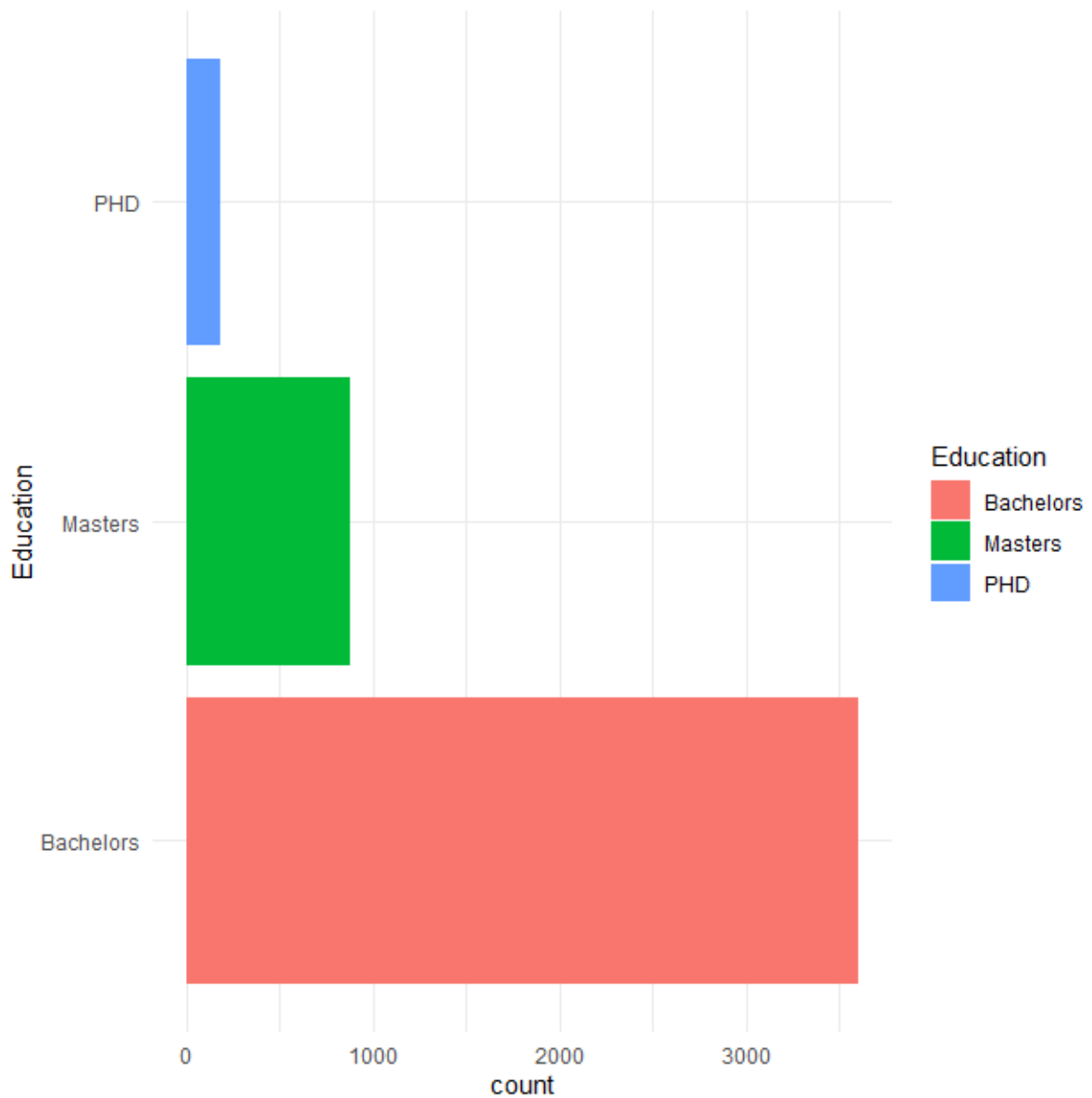
```
edu <- employee %>%  
  count(Education)
```

```
ggplot(edu,  
  aes(reorder(Education, n), n)) +  
  geom_segment(aes(xend = Education,  
    y = 0,  
    yend = n),  
    color = "gray") +  
  geom_point(size = 6,  
    color = "tomato") +  
  coord_flip() +  
  theme_minimal() +  
  labs(title = "Education Lollipop Chart",  
    x = "Education",  
    y = "Count")
```



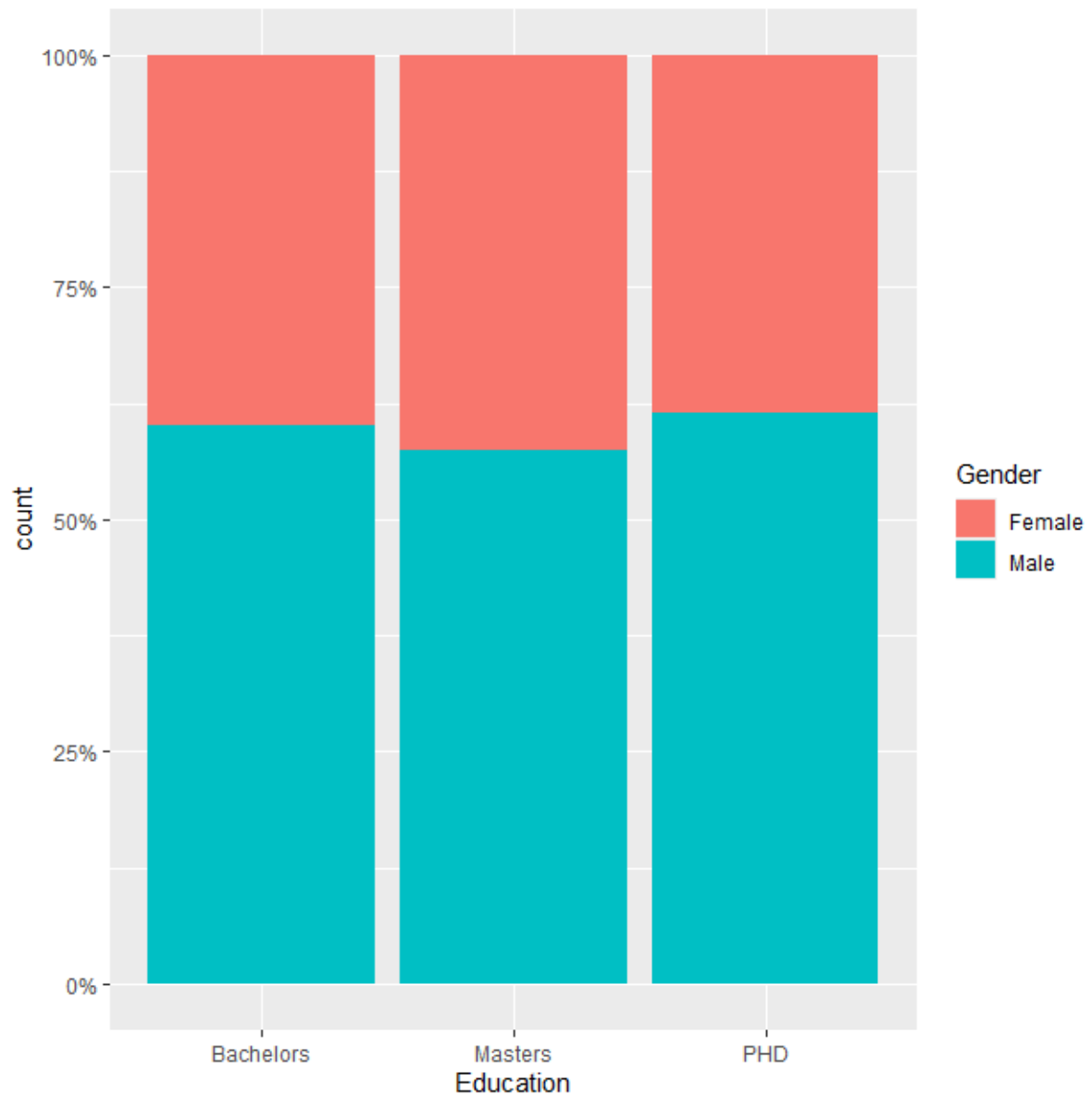
16. Horizontal Bar Chart

```
ggplot(employee, aes(Education, fill = Education)) +  
  geom_bar() +  
  coord_flip() +  
  theme_minimal()
```



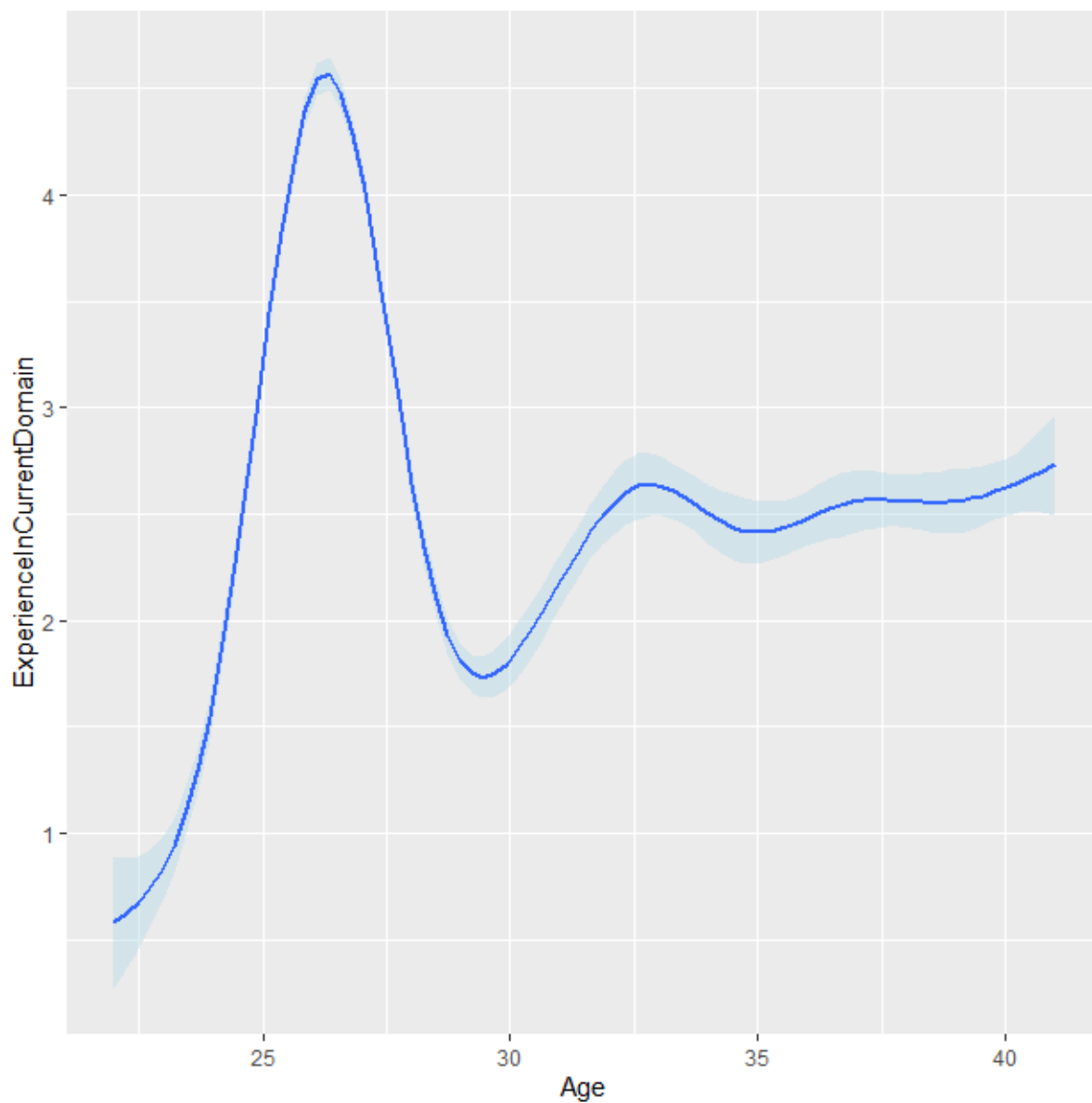
17. Percentage Stacked Bar Chart

```
ggplot(employee, aes(Education, fill = Gender)) +  
  geom_bar(position = "fill") +  
  scale_y_continuous(labels = scales::percent)
```



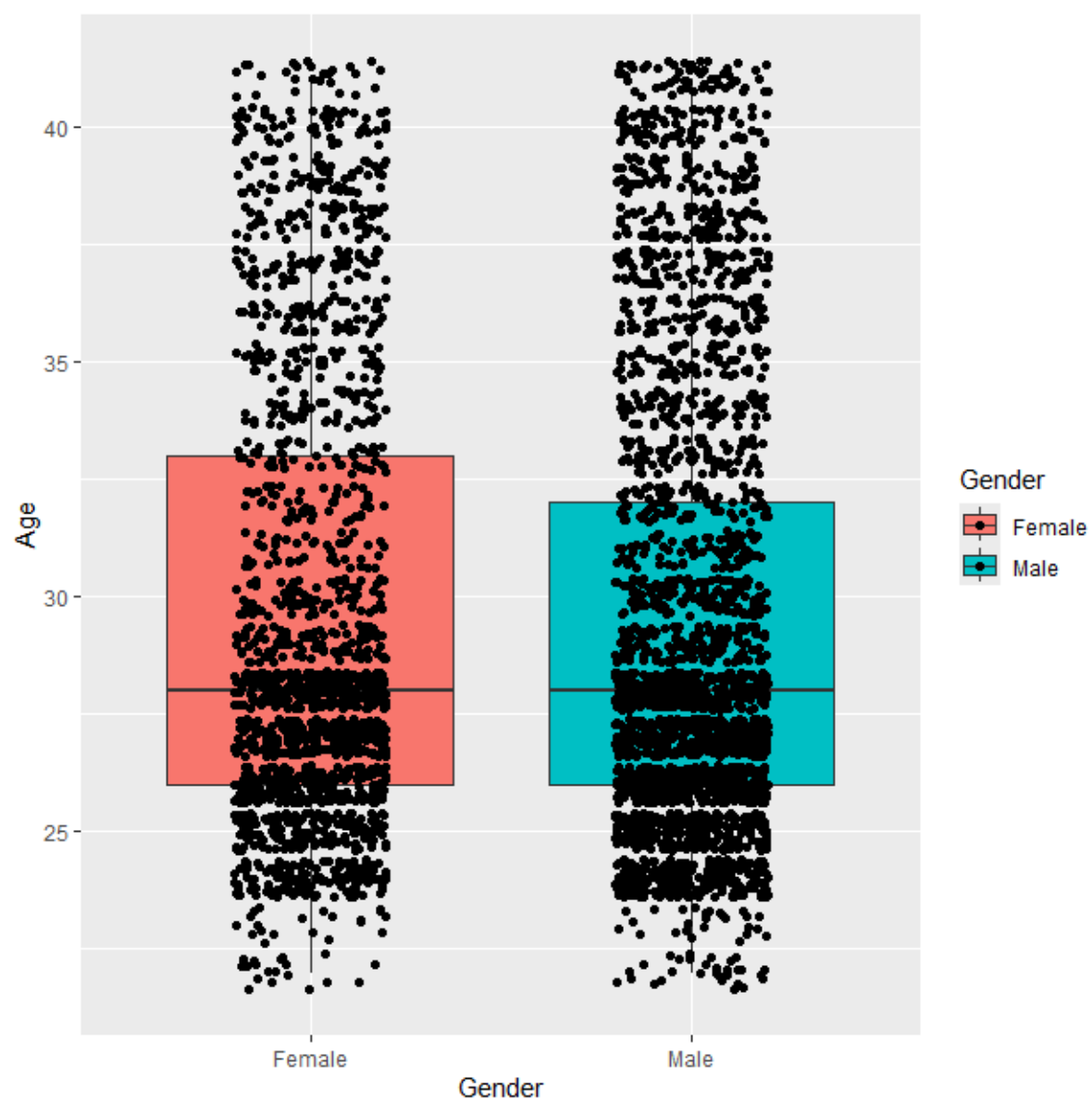
18. Smooth Line with Confidence Band

```
ggplot(employee,  
  aes(Age, ExperienceInCurrentDomain))+  
  geom_smooth(fill="lightblue")
```



19. Boxplot + Jitter

```
ggplot(employee,  
aes(Gender, Age, fill=Gender))+  
geom_boxplot()+  
geom_jitter(width=.2)
```



20. Histogram by Gender

```
ggplot(employee,  
aes(Age,fill=Gender))+  
geom_histogram(position="identity",  
alpha=.5)
```

